# №5

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2026-03-19

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# 1.1

- HTTP- Apache

# 1.1

- HTTP- Apache
- (HTTPS)

# 1.1

- HTTP- Apache
- (HTTPS)
- PHP

2

## 2.

- /etc/pki/tls/private

root@server:/etc/pki/tls/private

```
[user@server.user.net ~]$ sudo -i
[sudo] password for user:
[root@server.user.net ~]# mkdir -p /etc/pki/tls/private
[root@server.user.net ~]# ln -s /etc/pki/tls/private /etc/ssl/private
[root@server.user.net ~]# cd /etc/pki/tls/private
[root@server.user.net private]#
```

1: SSL- OpenSSL

- /etc/pki/tls/private
- openssl req -x509 -nodes -newkey rsa:2048

1: SSL- OpenSSL

- `/etc/pki/tls/private`
- `openssl req -x509 -nodes -newkey rsa:2048`
- DN (RU, Russia, Moscow, bahis, CN=bahis.net)

root@server:/etc/pki/tls/private

```
[user@server.user.net ~]$ sudo -i
[sudo] password for user:
[root@server.user.net ~]# mkdir -p /etc/pki/tls/private
[root@server.user.net ~]# ln -s /etc/pki/tls/private /etc/ssl/private
[root@server.user.net ~]# cd /etc/pki/tls/private
[root@server.user.net private]#
```

1: SSL- OpenSSL

- /etc/pki/tls/private
- openssl req -x509 -nodes -newkey rsa:2048
- DN (RU, Russia, Moscow, bahis, CN=bahis.net)
- /etc/ssl/certs

1: SSL- OpenSSL

- www.bahis.net.conf

```
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You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:RU
State or Province Name (full name) []:Russia
Locality Name (eg, city) [Default City]:Moscow
Organization Name (eg, company) [Default Company Ltd]:arsenio
Organizational Unit Name (eg, section) []:arsenio
Common Name (eg, your name or your server's hostname) []:user.net
Email Address []:user@user.net
[root@server.user.net private]#
[root@server.user.net private]#
```

2: Apache HTTPS

- www.bahis.net.conf
- <VirtualHost *:80> HTTPS (RewriteRule)

```
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-----
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There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:RU
State or Province Name (full name) []:Russia
Locality Name (eg, city) [Default City]:Moscow
Organization Name (eg, company) [Default Company Ltd]:arsenio
Organizational Unit Name (eg, section) []:arsenio
Common Name (eg, your name or your server's hostname) []:user.net
Email Address []:user@user.net
[root@server.user.net private]#
[root@server.user.net private]#
```

2: Apache HTTPS

- www.bahis.net.conf
- <VirtualHost *:80> HTTPS (RewriteRule)
- <VirtualHost *:443> SSLEngine on

```
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There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:RU
State or Province Name (full name) []:Russia
Locality Name (eg, city) [Default City]:Moscow
Organization Name (eg, company) [Default Company Ltd]:arsenio
Organizational Unit Name (eg, section) []:arsenio
Common Name (eg, your name or your server's hostname) []:user.net
Email Address []:user@user.net
[root@server.user.net private]#
[root@server.user.net private]#
```

2: Apache HTTPS

- www.bahis.net.conf
- <VirtualHost *:80> HTTPS (RewriteRule)
- <VirtualHost *:443> SSLEngine on
- SSLCertificateFile SSLCertificateKeyFile

```
.......+.............+..+......+.+......+.....+.........+.+..+............+......+.........+.+............+..+......+
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If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:RU
State or Province Name (full name) []:Russia
Locality Name (eg, city) [Default City]:Moscow
Organization Name (eg, company) [Default Company Ltd]:arsenio
Organizational Unit Name (eg, section) []:arsenio
Common Name (eg, your name or your server's hostname) []:user.net
Email Address []:user@user.net
[root@server.user.net private]#
[root@server.user.net private]#
```

2: Apache HTTPS

- www.bahis.net.conf
- <VirtualHost *:80> HTTPS (RewriteRule)
- <VirtualHost *:443> SSLEngine on
- SSLCertificateFile SSLCertificateKeyFile
- DocumentRoot

```
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Country Name (2 letter code) [XX]:RU
State or Province Name (full name) []:Russia
Locality Name (eg, city) [Default City]:Moscow
Organization Name (eg, company) [Default Company Ltd]:arsenio
Organizational Unit Name (eg, section) []:arsenio
Common Name (eg, your name or your server's hostname) []:user.net
Email Address []:user@user.net
[root@server.user.net private]#
[root@server.user.net private]#
```

2: Apache HTTPS

- firewalld

```
[user@server.user.net private]$
[user@server.user.net private]$ openssl req -x509 -nodes -newkey rsa:2048 -keyout www.user.net.key -out www.user.net
.crt              net private]$ openssl req -x509 -nodes -newkey rsa:2048 -keyout www.user.net.key -out
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...................+...+....+...........+....+..+....+.....+..........++++++
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++++++*......+........+...+...+....+...........+..........................+...+....+.........+......+......+..+.+.....
+..........+.....+....+...+........+...........+.......++++++
req: Can't open "www.user.net.key" for writing, Permission denied
[user@server.user.net private]$ sudo -i
[sudo] password for user:
[root@server.user.net ~]# openssl req -x509 -nodes -newkey rsa:2048 -keyout www.user.net.key -out
↪ www.user.net.crt
req: Option -out needs a value
req: Use -help for summary.
bash: ↪: command not found...
[root@server.user.net ~]# openssl req -x509 -nodes -newkey rsa:2048 -keyout www.user.net.key -out
↪ www.user.net.crt
req: Option -out needs a value
req: Use -help for summary.
bash: ↪: command not found...
[root@server.user.net ~]# cd /etc/pki/tls/private
[root@server.user.net private]# openssl req -x509 -nodes -newkey rsa:2048 -keyout www.user.net.key -out www.user.net
crt
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-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
```

- firewalld
- `https`

```
[user@server.user.net private]$
[user@server.user.net private]$ openssl req -x509 -nodes -newkey rsa:2048 -keyout www.user.net.key -out www.user.net
.crt              net private]$ openssl req -x509 -nodes -newkey rsa:2048 -keyout www.user.net.key -out
...........+......++++++
```

- firewalld
- `https`
-

```
[user@server.user.net private]$
[user@server.user.net private]$ openssl req -x509 -nodes -newkey rsa:2048 -keyout www.user.net.key -out www.user.net
.crt              net private]$ openssl req -x509 -nodes -newkey rsa:2048 -keyout www.user.net.key -out
...........+......++++++
```

- firewalld
- `https`
- 
- `httpd`

```
[user@server.user.net private]$
[user@server.user.net private]$ openssl req -x509 -nodes -newkey rsa:2048 -keyout www.user.net.key -out www.user.net
.crt              net private]$ openssl req -x509 -nodes -newkey rsa:2048 -keyout www.user.net.key -out
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++++++*......+........+...+...+....+...........+.........................+...+....+.........+......+......+..+.+.....
+..........+.....+....+...+........+...........+.......++++++
req: Can't open "www.user.net.key" for writing, Permission denied
[user@server.user.net private]$ sudo -i
[sudo] password for user:
[root@server.user.net ~]# openssl req -x509 -nodes -newkey rsa:2048 -keyout www.user.net.key -out
↪ www.user.net.crt
req: Option -out needs a value
req: Use -help for summary.
bash: ↪: command not found...
[root@server.user.net ~]# openssl req -x509 -nodes -newkey rsa:2048 -keyout www.user.net.key -out
↪ www.user.net.crt
req: Option -out needs a value
req: Use -help for summary.
bash: ↪: command not found...
[root@server.user.net ~]# cd /etc/pki/tls/private
[root@server.user.net private]# openssl req -x509 -nodes -newkey rsa:2048 -keyout www.user.net.key -out www.user.net
crt
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++++++
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
```

- https://www.bahis.net

root@server:/etc/httpd/conf.d

```
<VirtualHost *:80>
  ServerAdmin webmaster@user.net
  DocumentRoot /var/www/html/www.user.net
  ServerName www.user.net
  ServerAlias www.user.net
  ErrorLog logs/www.user.net-error_log
  CustomLog logs/www.user.net-access_log common
  RewriteEngine on
  RewriteRule ^(.*)$ https://%{HTTP_HOST}$1 [R=301,L]
 </VirtualHost>

<IfModule mod_ssl.c>
<VirtualHost *:443>
  SSLEngine on
  ServerAdmin webmaster@user.net
  DocumentRoot /var/www/html/www.user.net
  ServerName www.user.net
  ServerAlias www.user.net
  ErrorLog logs/www.user.net-error_log
  CustomLog logs/www.user.net-access_log common
  SSLCertificateFile /etc/ssl/certs/www.user.net.crt
  SSLCertificateKeyFile /etc/ssl/private/www.user.net.key
</VirtualHost>
</IfModule>
~
~
```

4: - HTTPS

- https://www.bahis.net
- HTTP → HTTPS

root@server:/etc/httpd/conf.d

```
<VirtualHost *:80>
  ServerAdmin webmaster@user.net
  DocumentRoot /var/www/html/www.user.net
  ServerName www.user.net
  ServerAlias www.user.net
  ErrorLog logs/www.user.net-error_log
  CustomLog logs/www.user.net-access_log common
  RewriteEngine on
  RewriteRule ^(.*)$ https://%{HTTP_HOST}$1 [R=301,L]
 </VirtualHost>

<IfModule mod_ssl.c>
<VirtualHost *:443>
  SSLEngine on
  ServerAdmin webmaster@user.net
  DocumentRoot /var/www/html/www.user.net
  ServerName www.user.net
  ServerAlias www.user.net
  ErrorLog logs/www.user.net-error_log
  CustomLog logs/www.user.net-access_log common
  SSLCertificateFile /etc/ssl/certs/www.user.net.crt
  SSLCertificateKeyFile /etc/ssl/private/www.user.net.key
</VirtualHost>
</IfModule>
~
~
```

4: - HTTPS

- https://www.bahis.net
- HTTP → HTTPS
-

root@server:/etc/httpd/conf.d

```
<VirtualHost *:80>
  ServerAdmin webmaster@user.net
  DocumentRoot /var/www/html/www.user.net
  ServerName www.user.net
  ServerAlias www.user.net
  ErrorLog logs/www.user.net-error_log
  CustomLog logs/www.user.net-access_log common
  RewriteEngine on
  RewriteRule ^(.*)$ https://%{HTTP_HOST}$1 [R=301,L]
 </VirtualHost>

<IfModule mod_ssl.c>
<VirtualHost *:443>
  SSLEngine on
  ServerAdmin webmaster@user.net
  DocumentRoot /var/www/html/www.user.net
  ServerName www.user.net
  ServerAlias www.user.net
  ErrorLog logs/www.user.net-error_log
  CustomLog logs/www.user.net-access_log common
  SSLCertificateFile /etc/ssl/certs/www.user.net.crt
  SSLCertificateKeyFile /etc/ssl/private/www.user.net.key
</VirtualHost>
</IfModule>
~
~
```

4: - HTTPS

- https://www.bahis.net
- HTTP → HTTPS
- 
- HTTPS

root@server:/etc/httpd/conf.d

```
<VirtualHost *:80>
  ServerAdmin webmaster@user.net
  DocumentRoot /var/www/html/www.user.net
  ServerName www.user.net
  ServerAlias www.user.net
  ErrorLog logs/www.user.net-error_log
  CustomLog logs/www.user.net-access_log common
  RewriteEngine on
  RewriteRule ^(.*)$ https://%{HTTP_HOST}$1 [R=301,L]
 </VirtualHost>

<IfModule mod_ssl.c>
<VirtualHost *:443>
  SSLEngine on
  ServerAdmin webmaster@user.net
  DocumentRoot /var/www/html/www.user.net
  ServerName www.user.net
  ServerAlias www.user.net
  ErrorLog logs/www.user.net-error_log
  CustomLog logs/www.user.net-access_log common
  SSLCertificateFile /etc/ssl/certs/www.user.net.crt
  SSLCertificateKeyFile /etc/ssl/private/www.user.net.key
</VirtualHost>
</IfModule>
~
~
```

4: - HTTPS

# 2.2 HTTP- PHP

- php dnf

```
[root@server.user.net ~]# dnf -y install php
Last metadata expiration check: 0:15:28 ago on Fri 06 Mar 2026 07:45:45 PM UTC.
Dependencies resolved.
================================================================================================================
 Package                        Architecture         Version                         Repository              Size
================================================================================================================
Installing:
 php                            x86_64               8.0.30-5.el9_7                  appstream               8.8 k
Installing dependencies:
 nginx-filesystem               noarch               2:1.20.1-24.el9                 appstream               9.2 k
 php-common                     x86_64               8.0.30-5.el9_7                  appstream               667 k
Installing weak dependencies:
 php-cli                        x86_64               8.0.30-5.el9_7                  appstream               3.1 M
 php-fpm                        x86_64               8.0.30-5.el9_7                  appstream               1.6 M
 php-mbstring                   x86_64               8.0.30-5.el9_7                  appstream               468 k
 php-opcache                    x86_64               8.0.30-5.el9_7                  appstream               510 k
 php-pdo                        x86_64               8.0.30-5.el9_7                  appstream                82 k
 php-xml                        x86_64               8.0.30-5.el9_7                  appstream               133 k

Transaction Summary
================================================================================================================
Install  9 Packages

Total download size: 6.5 M
Installed size: 35 M
Downloading Packages:
(1/9): nginx-filesystem-1.20.1-24.el9.noarch.rpm                                 36 kB/s | 9.2 kB     00:00
(2/9): php-8.0.30-5.el9_7.x86_64.rpm                                             28 kB/s | 8.8 kB     00:00
(3/9): php-common-8.0.30-5.el9_7.x86_64.rpm                                     188 kB/s | 667 kB     00:03
(4/9): php-fpm-8.0.30-5.el9_7.x86_64.rpm                                        315 kB/s | 1.6 MB     00:05
(5/9): php-mbstring-8.0.30-5.el9_7.x86_64.rpm                                   133 kB/s | 468 kB     00:03
```

# 2.2 HTTP- PHP

- php dnf
- PHP 8.0.30

```
[root@server.user.net ~]# dnf -y install php
Last metadata expiration check: 0:15:28 ago on Fri 06 Mar 2026 07:45:45 PM UTC.
Dependencies resolved.
================================================================================================
 Package                        Architecture        Version                        Repository          Size
================================================================================================
Installing:
 php                            x86_64              8.0.30-5.el9_7                 appstream           8.8 k
Installing dependencies:
 nginx-filesystem               noarch              2:1.20.1-24.el9                appstream           9.2 k
 php-common                     x86_64              8.0.30-5.el9_7                 appstream           667 k
Installing weak dependencies:
 php-cli                        x86_64              8.0.30-5.el9_7                 appstream           3.1 M
 php-fpm                        x86_64              8.0.30-5.el9_7                 appstream           1.6 M
 php-mbstring                   x86_64              8.0.30-5.el9_7                 appstream           468 k
 php-opcache                    x86_64              8.0.30-5.el9_7                 appstream           510 k
 php-pdo                        x86_64              8.0.30-5.el9_7                 appstream            82 k
 php-xml                        x86_64              8.0.30-5.el9_7                 appstream           133 k

Transaction Summary
================================================================================================
Install  9 Packages

Total download size: 6.5 M
Installed size: 35 M
Downloading Packages:
(1/9): nginx-filesystem-1.20.1-24.el9.noarch.rpm                 36 kB/s | 9.2 kB     00:00
(2/9): php-8.0.30-5.el9_7.x86_64.rpm                             28 kB/s | 8.8 kB     00:00
(3/9): php-common-8.0.30-5.el9_7.x86_64.rpm                     188 kB/s | 667 kB     00:03
(4/9): php-fpm-8.0.30-5.el9_7.x86_64.rpm                        315 kB/s | 1.6 MB     00:05
(5/9): php-mbstring-8.0.30-5.el9_7.x86_64.rpm                   133 kB/s | 468 kB     00:03
```

# 2.2 HTTP- PHP

- `php` `dnf`
- PHP 8.0.30
-

```
[root@server.user.net ~]# dnf -y install php
Last metadata expiration check: 0:15:28 ago on Fri 06 Mar 2026 07:45:45 PM UTC.
Dependencies resolved.
================================================================================================================
 Package                        Architecture         Version                         Repository              Size
================================================================================================================
Installing:
 php                            x86_64               8.0.30-5.el9_7                  appstream               8.8 k
Installing dependencies:
 nginx-filesystem               noarch               2:1.20.1-24.el9                 appstream               9.2 k
 php-common                     x86_64               8.0.30-5.el9_7                  appstream               667 k
Installing weak dependencies:
 php-cli                        x86_64               8.0.30-5.el9_7                  appstream               3.1 M
 php-fpm                        x86_64               8.0.30-5.el9_7                  appstream               1.6 M
 php-mbstring                   x86_64               8.0.30-5.el9_7                  appstream               468 k
 php-opcache                    x86_64               8.0.30-5.el9_7                  appstream               510 k
 php-pdo                        x86_64               8.0.30-5.el9_7                  appstream                82 k
 php-xml                        x86_64               8.0.30-5.el9_7                  appstream               133 k

Transaction Summary
================================================================================================================
Install  9 Packages

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(2/9): php-8.0.30-5.el9_7.x86_64.rpm                                             28 kB/s | 8.8 kB     00:00
(3/9): php-common-8.0.30-5.el9_7.x86_64.rpm                                     188 kB/s | 667 kB     00:03
(4/9): php-fpm-8.0.30-5.el9_7.x86_64.rpm                                        315 kB/s | 1.6 MB     00:05
(5/9): php-mbstring-8.0.30-5.el9_7.x86_64.rpm                                   133 kB/s | 468 kB     00:03
```

- index.html

```
[root@server.user.net ~]#
[root@server.user.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https imap imaps mountd nfs ntp pop3 pop3s rpc-bind samba smtp smtp-submission s
sh ssh-custom
[root@server.user.net ~]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-k5-client amanda-k5-client amqp amqps apcupsd audit ausweisapp2 bacula
 bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-t
estnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit collectd condor-collector crate
db ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls docker-registry docker-swarm
 dropbox-lansync elasticsearch etcd-client etcd-server finger foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-l
daps freeipa-replication freeipa-trust ftp galera ganglia-client ganglia-master git gpsd grafana gre high-availabili
ty http http3 https ident imap imaps ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect
kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kub
e-controller-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube
-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client
llmnr-tcp llmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur
mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovirt-storagecon
sole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-e
xporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-sentinel rpc-bind rq
uotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmp
tls snmptls-trap snmptrap spideroak-lansync spotify-sync squid ssdp ssh ssh-custom steam-streaming svdrp svn syncthi
ng syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks transmission-c
lient upnp-client vdsm vnc-server warpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-disc
overy-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-serve
r zerotier
[root@server.user.net ~]# firewall-cmd --add-service=https
Warning: ALREADY_ENABLED: 'https' already in 'public'
success
[root@server.user.net ~]# firewall-cmd --add-service=https --permanent
Warning: ALREADY_ENABLED: https
success
[root@server.user.net ~]# firewall-cmd --reload
```

- index.html
- index.php phpinfo();

```
[root@server.user.net ~]#
[root@server.user.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https imap imaps mountd nfs ntp pop3 pop3s rpc-bind samba smtp smtp-submission s
sh ssh-custom
[root@server.user.net ~]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-k5-client amqp amqps apcupsd audit ausweisapp2 bacula
 bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-t
estnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit collectd condor-collector crate
db ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls docker-registry docker-swarm
 dropbox-lansync elasticsearch etcd-client etcd-server finger foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-l
daps freeipa-replication freeipa-trust ftp galera ganglia-client ganglia-master git gpsd grafana gre high-availabili
ty http http3 https ident imap imaps ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect
kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kub
e-controller-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube
-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client
llmnr-tcp llmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur
mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovirt-storagecon
sole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-e
xporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-sentinel rpc-bind rq
uotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmp
tls snmptls-trap snmptrap spideroak-lansync spotify-sync squid ssdp ssh ssh-custom steam-streaming svdrp svn syncthi
ng syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks transmission-c
lient upnp-client vdsm vnc-server warpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-disc
overy-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-serve
r zerotier
[root@server.user.net ~]# firewall-cmd --add-service=https
Warning: ALREADY_ENABLED: 'https' already in 'public'
success
[root@server.user.net ~]# firewall-cmd --add-service=https --permanent
Warning: ALREADY_ENABLED: https
success
[root@server.user.net ~]# firewall-cmd --reload
```

- index.html
- index.php phpinfo();
- chown -R apache:apache /var/www

```
[root@server.user.net ~]#
[root@server.user.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https imap imaps mountd nfs ntp pop3 pop3s rpc-bind samba smtp smtp-submission s
sh ssh-custom
[root@server.user.net ~]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisapp2 bacula
 bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-t
estnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit collectd condor-collector crate
db ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls docker-registry docker-swarm
 dropbox-lansync elasticsearch etcd-client etcd-server finger foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-l
daps freeipa-replication freeipa-trust ftp galera ganglia-client ganglia-master git gpsd grafana gre high-availabili
ty http http3 https ident imap imaps ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect
kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kub
e-controller-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube
-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client
llmnr-tcp llmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur
mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovirt-storagecon
sole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-e
xporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-sentinel rpc-bind rq
uotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmp
tls snmptls-trap snmptrap spideroak-lansync spotify-sync squid ssdp ssh ssh-custom steam-streaming svdrp svn syncthi
ng syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks transmission-c
lient upnp-client vdsm vnc-server warpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-disc
overy-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-serve
r zerotier
[root@server.user.net ~]# firewall-cmd --add-service=https
Warning: ALREADY_ENABLED: 'https' already in 'public'
success
[root@server.user.net ~]# firewall-cmd --add-service=https --permanent
Warning: ALREADY_ENABLED: https
success
[root@server.user.net ~]# firewall-cmd --reload
```

- index.html
- index.php phpinfo();
- chown -R apache:apache /var/www
- SELinux-

```
[root@server.user.net ~]#
[root@server.user.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https imap imaps mountd nfs ntp pop3 pop3s rpc-bind samba smtp smtp-submission s
sh ssh-custom
[root@server.user.net ~]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisapp2 bacula
 bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-t
estnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit collectd condor-collector crate
db ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls docker-registry docker-swarm
 dropbox-lansync elasticsearch etcd-client etcd-server finger foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-l
daps freeipa-replication freeipa-trust ftp galera ganglia-client ganglia-master git gpsd grafana gre high-availabili
ty http http3 https ident imap imaps ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect
kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kub
e-controller-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube
-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client
llmnr-tcp llmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur
mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovirt-storagecon
sole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-e
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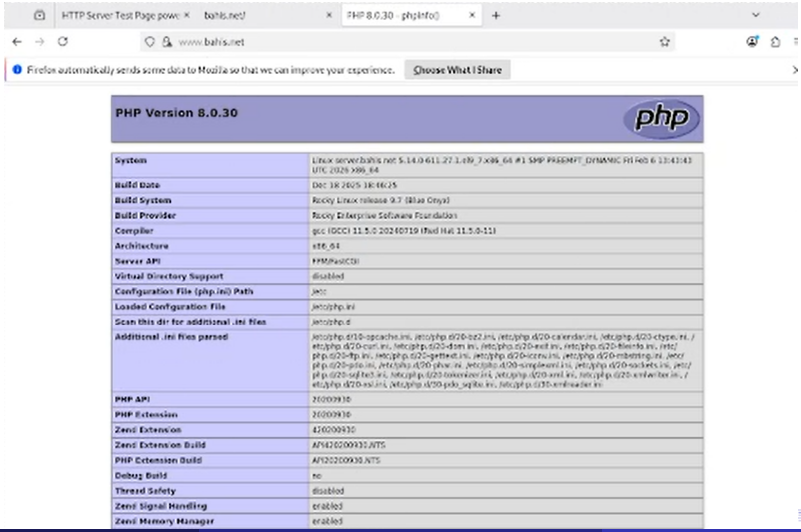
overy-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-serve
r zerotier
[root@server.user.net ~]# firewall-cmd --add-service=https
Warning: ALREADY_ENABLED: 'https' already in 'public'
success
[root@server.user.net ~]# firewall-cmd --add-service=https --permanent
Warning: ALREADY_ENABLED: https
success
[root@server.user.net ~]# firewall-cmd --reload
```

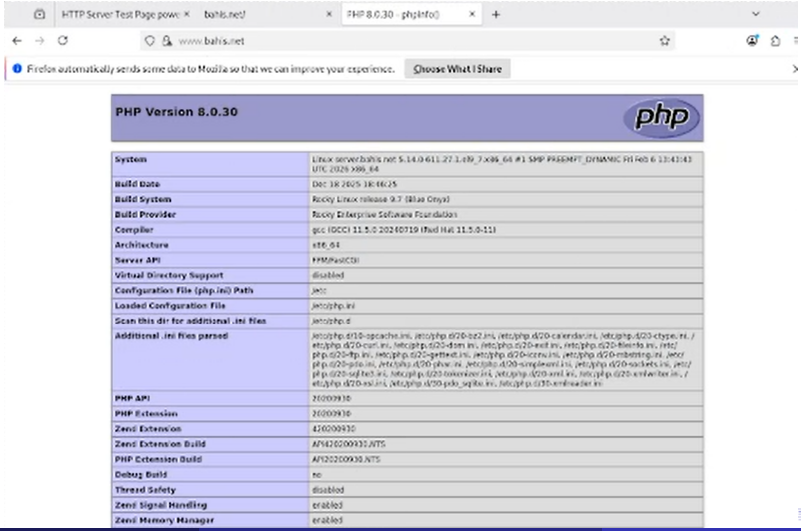
- index.html
- index.php phpinfo();
- chown -R apache:apache /var/www
- SELinux-
- httpd

```
[root@server.user.net ~]#
[root@server.user.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https imap imaps mountd nfs ntp pop3 pop3s rpc-bind samba smtp smtp-submission s
sh ssh-custom
[root@server.user.net ~]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisapp2 bacula
 bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-t
estnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit collectd condor-collector crate
db ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls docker-registry docker-swarm
 dropbox-lansync elasticsearch etcd-client etcd-server finger foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-l
daps freeipa-replication freeipa-trust ftp galera ganglia-client ganglia-master git gpsd grafana gre high-availabili
ty http http3 https ident imap imaps ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect
kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kub
e-controller-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube
-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client
llmnr-tcp llmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur
mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovirt-storagecon
sole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-e
xporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-sentinel rpc-bind rq
uotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmp
tls snmptls-trap snmptrap spideroak-lansync spotify-sync squid ssdp ssh ssh-custom steam-streaming svdrp svn syncthi
ng syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks transmission-c
lient upnp-client vdsm vnc-server warpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-disc
overy-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-serve
r zerotier
[root@server.user.net ~]# firewall-cmd --add-service=https
Warning: ALREADY_ENABLED: 'https' already in 'public'
success
[root@server.user.net ~]# firewall-cmd --add-service=https --permanent
Warning: ALREADY_ENABLED: https
success
[root@server.user.net ~]# firewall-cmd --reload
```

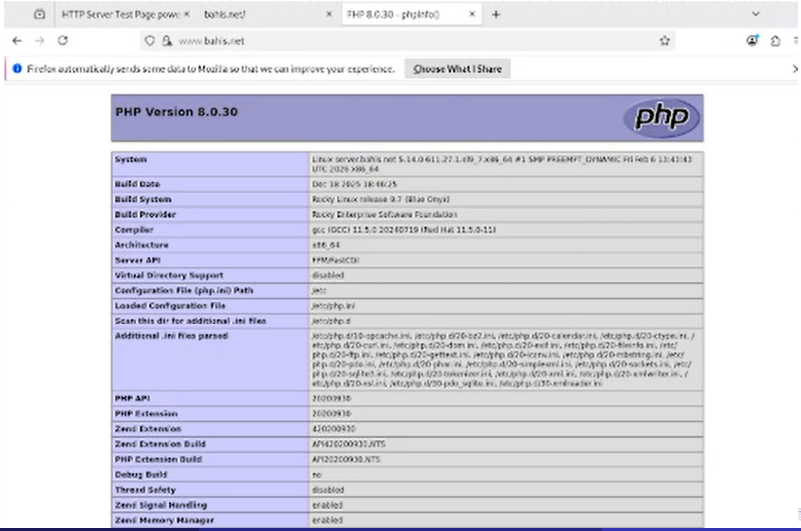
- www.bahis.net



- www.bahis.net
- phpinfo()



- www.bahis.net
- phpinfo()
- PHP Apache



- /etc/httpd/conf.d

```
[root@server.user.net ~]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http/etc/httpd/conf.d
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/autoindex.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/fcgid.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/manual.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/README'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/rocky-snipolicy.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/userdir.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/welcome.conf'?
[root@server.user.net ~]#
[root@server.user.net ~]#
[root@server.user.net ~]# cp -R /var/www/html/* /vagrant/provision/server/http/var/www/html
cp: overwrite '/vagrant/provision/server/http/var/www/html/www.user.net/index.html'?
[root@server.user.net ~]#
[root@server.user.net ~]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/private
[root@server.user.net ~]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/certs
[root@server.user.net ~]# cp -R /etc/pki/tls/private/www.user.net.key /vagrant/provision/server/http/etc/pki/tls/pri
vate
[root@server.user.net ~]# cp -R /etc/pki/tls/certs/www.user.net.crt /vagrant/provision/server/http/etc/pki/tls/certs
cp: cannot stat '/etc/pki/tls/certs/www.user.net.crt': No such file or directory
[root@server.user.net ~]# cp -R /etc/pki/tls/certs/www.user.net.crt  /vagrant/provision/server/http/etc/pki/tls/cert
s
```

8: HTTP SSL provision

# 2.3

- /etc/httpd/conf.d
- - /var/www/html

```
[root@server.user.net ~]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http/etc/httpd/conf.d
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/autoindex.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/fcgid.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/manual.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/README'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/rocky-snipolicy.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/userdir.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/welcome.conf'?
[root@server.user.net ~]#
[root@server.user.net ~]#
[root@server.user.net ~]# cp -R /var/www/html/* /vagrant/provision/server/http/var/www/html
cp: overwrite '/vagrant/provision/server/http/var/www/html/www.user.net/index.html'?
[root@server.user.net ~]#
[root@server.user.net ~]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/private
[root@server.user.net ~]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/certs
[root@server.user.net ~]# cp -R /etc/pki/tls/private/www.user.net.key /vagrant/provision/server/http/etc/pki/tls/pri
vate
[root@server.user.net ~]# cp -R /etc/pki/tls/certs/www.user.net.crt /vagrant/provision/server/http/etc/pki/tls/certs
cp: cannot stat '/etc/pki/tls/certs/www.user.net.crt': No such file or directory
[root@server.user.net ~]# cp -R /etc/pki/tls/certs/www.user.net.crt  /vagrant/provision/server/http/etc/pki/tls/cert
s
```

8: HTTP SSL provision

# 2.3

- /etc/httpd/conf.d
- - /var/www/html
- provision

```
[root@server.user.net ~]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http/etc/httpd/conf.d
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/autoindex.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/fcgid.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/manual.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/README'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/rocky-snipolicy.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/userdir.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/welcome.conf'?
[root@server.user.net ~]#
[root@server.user.net ~]#
[root@server.user.net ~]# cp -R /var/www/html/* /vagrant/provision/server/http/var/www/html
cp: overwrite '/vagrant/provision/server/http/var/www/html/www.user.net/index.html'?
[root@server.user.net ~]#
[root@server.user.net ~]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/private
[root@server.user.net ~]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/certs
[root@server.user.net ~]# cp -R /etc/pki/tls/private/www.user.net.key /vagrant/provision/server/http/etc/pki/tls/pri
vate
[root@server.user.net ~]# cp -R /etc/pki/tls/certs/www.user.net.crt /vagrant/provision/server/http/etc/pki/tls/certs
cp: cannot stat '/etc/pki/tls/certs/www.user.net.crt': No such file or directory
[root@server.user.net ~]# cp -R /etc/pki/tls/certs/www.user.net.crt  /vagrant/provision/server/http/etc/pki/tls/cert
s
```

8: HTTP SSL provision

# 2.3

- /etc/httpd/conf.d
- - /var/www/html
- provision
- www.bahis.net.key www.bahis.net.crt

```
[root@server.user.net ~]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http/etc/httpd/conf.d
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/autoindex.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/fcgid.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/manual.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/README'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/rocky-snipolicy.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/userdir.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/welcome.conf'?
[root@server.user.net ~]#
[root@server.user.net ~]#
[root@server.user.net ~]# cp -R /var/www/html/* /vagrant/provision/server/http/var/www/html
cp: overwrite '/vagrant/provision/server/http/var/www/html/www.user.net/index.html'?
[root@server.user.net ~]#
[root@server.user.net ~]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/private
[root@server.user.net ~]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/certs
[root@server.user.net ~]# cp -R /etc/pki/tls/private/www.user.net.key /vagrant/provision/server/http/etc/pki/tls/pri
vate
[root@server.user.net ~]# cp -R /etc/pki/tls/certs/www.user.net.crt /vagrant/provision/server/http/etc/pki/tls/certs
cp: cannot stat '/etc/pki/tls/certs/www.user.net.crt': No such file or directory
[root@server.user.net ~]# cp -R /etc/pki/tls/certs/www.user.net.crt  /vagrant/provision/server/http/etc/pki/tls/cert
s
```

8: HTTP SSL provision

- http.sh php

```
GNU nano 5.6.1                    /vagrant/provision/server/http.s
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y groupinstall "Basic Web Server"
echo "Copy configuration files"
cp -R /vagrant/provision/server/http/etc/httpd/* /etc/httpd
cp -R /vagrant/provision/server/http/var/www/* /var/www
chown -R apache:apache /var/www
restorecon -vR /etc
restorecon -vR /var/www
echo "Configure firewall"
firewall-cmd --add-service=http
firewall-cmd --add-service=http --permanent
echo "Start http service"
systemctl enable httpd
systemctl start httpd
```

- http.sh php
- /etc/httpd /var/www

```
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- http.sh  php
- /etc/httpd  /var/www
- apache

```
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- http.sh  php
- /etc/httpd  /var/www
- apache
- SELinux-

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- http.sh  php
- /etc/httpd  /var/www
- apache
- SELinux-
- firewalld  http  https

```
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- http.sh  php
- /etc/httpd  /var/www
- apache
- SELinux-
- firewalld  http  https
- httpd

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# 3

## 3.

# 3.1

- HTTPS

# 3.1

- HTTPS
- HTTP $\rightarrow$ HTTPS

# 3.1

- HTTPS
- HTTP $\rightarrow$ HTTPS
- `https` firewall

# 3.1

- HTTPS
- HTTP $\rightarrow$ HTTPS
- `https` firewall
- PHP Apache

# 3.1

- HTTPS
- HTTP → HTTPS
- `https` firewall
- PHP Apache
- PHP- (`phpinfo()`)

# 3.1

- HTTPS
- HTTP → HTTPS
- `https` firewall
- PHP Apache
- PHP- (`phpinfo()`)
- provisioning-

# 3.1

- HTTPS
- HTTP → HTTPS
- `https` firewall
- PHP Apache
- PHP- (`phpinfo()`)
- provisioning-
- `http.sh`